CHAPTER C20
SITE CLASSIFICATION PROCEDURE FOR SEISMIC DESIGN

C20.1 SITE CLASSIFICATION

Site classification procedures are given in Chapter 20 for the purpose of classifying the site and determining site coefficients and site-adjusted risk-targeted maximum considered earthquake ground motions in accordance with Section 11.4.3. Site classification procedures are also used to define the site conditions for which site-specific site response analyses are required to obtain site ground motions in accordance with Section 11.4.7 and Chapter 21.

C20.3 SITE CLASS DEFINITIONS

C20.3.1 Site Class F. Site Class F conditions are conditions for which the site coefficients F_a and F_v in Tables 11.4-1 and 11.4-2 may not be applicable for site response analyses required by Section 11.4.7; they are defined in this section. For three of the categories of Site Class F soils—Category 1 liquefiable soils, Category 3 very high plasticity clays, and Category 4 very thick soft/medium stiff clays—exceptions to the requirement to conduct site response analyses are given, provided that certain conditions and requirements are satisfied. These exceptions are discussed below.

Category 1. For liquefiable soils in Category 1, an exception to conducting site response analyses was developed by Technical Subcommittee 3, Foundations and Geotechnical Considerations, of the BSSC Provisions Update Committee and was first published in the 2000 NEHRP Provisions (FEMA 2001). The exception is made for short-period structures, defined for purposes of the exception as having fundamental periods of vibration equal to or less than 0.5 s. For such structures, it is permissible to determine site coefficients F_a and F_v from Tables 11.4-1 and 11.4-2 assuming that liquefaction does not occur because ground motion data obtained in liquefied soil areas during earthquakes indicate that short-period ground motions are generally reduced in amplitude because of liquefaction, whereas long-period ground motions may be amplified by liquefaction. Note, however, that this exception does not affect the requirement in Section 11.8 to assess liquefaction potential as a geologic hazard and develop hazard mitigation measures if required.

Categories 3 and 4. For very high plasticity clays in Category 3 and very thick soft/medium stiff clays in Category 4, exceptions to conducting site response analysis were evaluated and developed by an ad hoc geotechnical task committee formed to support the BSSC Simplified Seismic Project in 2011 and 2012. Exceptions for Categories 3 and 4 were limited to sites of expected low ground motions, i.e., Seismic Design Category B (SDC B) as defined in Tables 11.6-1 and 11.6-2.

For Category 3 very high plasticity clays, the task committee evaluated published research and conducted supplemental analyses to evaluate effects of increasing soil plasticity on soil amplification. From these evaluations, exceptions were developed for scaling the F_a and F_v site coefficients upward from the values given in Tables 11.4-1 and 11.4-2 for Site Class D or E by factors that are a function of soil plasticity as quantified by soil plasticity index (PI).

For Category 4 very thick soft/medium stiff clays, the task committee reviewed analyses conducted for a 1992 Site Response Workshop (Dobry et al. 2000) (at which recommendations were developed for site classifications and site coefficients that have been in the ASCE 7 standard through ASCE 7-10) and conducted supplemental analyses to evaluate effects of clay thickness on soil amplification. These analyses indicated that maximum amplifications should be insensitive to soil thickness and that site coefficients F_a and F_v for Site Class E should be adequate or conservative in most cases for soft/medium stiff clays thicker than 120 ft (36.576 m). A supplemental analysis indicated that soft/medium stiff clay thicknesses greater than 120 ft (36.576 m) would be uncommon for sites in the United States because of the higher overburden pressures at these depths that would generally result in clays being in a category of “stiff” rather than “soft to medium stiff.” Clays meeting the Category 4 definition of Site Class F have undrained shear strengths, s_u, less than 1,000 psf (47.9 kN/m^2), whereas stiff clays have higher s_u values.

Sections C20.3.2 through C20.3.5. These sections and Table 20.3-1 provide definitions for Site Classes A through E. Except for the additional definitions for Site Class E in Section 20.3.2, the site classes are defined fundamentally in terms of the average small-strain shear wave velocity in the top 100 ft (30 m) of the soil or rock profile. If shear wave velocities are available for the site, they should be used to classify the site. However, recognizing that in many cases shear wave velocities are not available for the site, alternative definitions of the site classes also are included. These definitions are based on geotechnical parameters: standard penetration resistance for cohesionless soils and rock, and standard penetration resistance and undrained shear strength for cohesive soils. The alternative definitions are intended to be conservative because the correlation between site coefficients and these geotechnical parameters is more uncertain than the correlation with shear wave velocity. That is, values of F_a and F_v tend to be smaller if the site class is based on shear wave velocity rather than on the geotechnical parameters. Also, the site class definitions should not be interpreted as implying any specific numerical correlation between shear wave velocity and standard penetration resistance or undrained shear strength.

Although the site class definitions in Sections 20.3.2 through 20.3.5 are straightforward, there are aspects of these assessments that may require additional judgment and interpretation. Highly variable subsurface conditions beneath a building footprint could result in overly conservative or unconservative site classification. Isolated soft soil layers within an otherwise firm soil site may not affect the overall site response if the predominant soil conditions do not include such strata. Conversely, site response studies have shown that continuous, thin, soft clay strata may affect the site amplification.

The site class should reflect the soil conditions that affect the ground motion input to the structure or a significant portion of the structure. For structures that receive substantial ground motion input from shallow soils (for example, structures with shallow spread footings, with laterally flexible piles, or with basements where substantial ground motion input to the structure may come through the sidewalls), it is reasonable to classify the site on the basis of the top 100 ft (30 m) of soils below the ground surface. Conversely, for structures with basements supported on firm soils or rock below soft soils, it may be reasonable to classify the site on the basis of the soils or rock below the mat, if it can be justified that the soft soils contribute very little to the response of the structure.

Buildings on sites with sloping bedrock or highly variable soil deposits across the building area require careful study because the input motion may vary across the building (for example, if a portion of the building is on rock and the rest is over weak soils). Site-specific studies including two- or three-dimensional modeling may be used in such cases to evaluate the subsurface conditions and site and superstructure response. Other conditions that may warrant site-specific evaluation include the presence of low shear wave velocity soils below a depth of 100 ft (30 m), location of the site in a sedimentary basin, or subsurface or topographic conditions with strong two- and three-dimensional site response effects. Individuals with appropriate expertise in seismic ground motions should participate in evaluations of the need for and nature of such site-specific studies.

C20.4 DEFINITIONS OF SITE CLASS PARAMETERS

Section 20.4 provides formulas for defining site classes in accordance with definitions in Section 20.3 and Table 20.3-1. Eq. (20.4-1) is for determining the effective average small-strain shear wave velocity, $\bar{v}_s$, to a depth of 100 ft (30 m) at a site. This equation defines $\bar{v}_s$ as 100 ft (30 m) divided by the sum of the times for a shear wave to travel through each layer within the upper 100 ft (30 m), where travel time for each layer is calculated as the layer thickness divided by the small-strain shear wave velocity for the layer. It is important that this method of averaging be used because it may result in a significantly lower effective average shear wave velocity than the velocity that would be obtained by directly averaging the velocities of the individual layers.

For example, consider a soil profile that has four 25-ft (7.62-m) thick layers with shear wave velocities of 500 ft/s (152.4 m/s), 1,000 ft/s (304.8 m/s), 1,500 ft/s (457.2 m/s), and 2,000 ft/s (609.2 m/s). The arithmetic average of the shear wave velocities is 1,250 ft/s (381.0 m/s) (corresponding to Site Class C), but Eq. (20.4-1) produces a value of 960 ft/s (292.6 m/s) (corresponding to Site Class D). The Eq. (20.4-1) value is appropriate because the four layers are being represented by one layer with the same wave passage time.

Eq. (20.4-2) is for classifying the site using the average standard penetration resistance blow count, $\bar{N}$, for cohesionless soils, cohesive soils, and rock in the upper 100 ft (30 m). A method of averaging analogous to the method of Eq. (20.4-1) for shear wave velocity is used. The maximum value of N that may be used for any depth of measurement in soil or rock is 100 blows/ft (305 blows/m). For the common situation where rock is encountered, the standard penetration resistance, N, for rock layers is taken as 100.

Eqs. (20.4-3) and (20.4-4) are for classifying the site using the standard penetration resistance of cohesionless soil layers, $\bar{N}_{ch}$, and the undrained shear strength of cohesive soil layers, $\bar{s}_u$, within the top 100 ft (30 m). These equations are provided as an alternative to using Eq. (20.4-2), for which N values in all geologic materials in the top 100 ft (30 m) are used. Where using Eqs. (20.4-3) and (20.4-4), only the respective thicknesses of cohesionless soils and cohesive soils within the top 100 ft (30 m) are used.

REFERENCES

Dobry, R., Borcherdt, R. D., Crouse, C. B., Idriss, I. M., Joyner, W. B., Martin, G. R., et al. (2000). “New site coefficients and site classification system used in recent building seismic code provisions.” *Earthq. Spectra*, 16(1), 41–67.

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